PHYSIGYM: Bridging Agent-Based Biological Simulation and Reinforcement Learning A State-Space Study on Tumor Microenvironment Control

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Abstract

This paper presents PHYSIGYM, a framework that integrates agent-based biological simulation within standardised reinforcement learning environments. By connecting the agent-based modelling framework PHYSICELL with the Gymnasium API, we provide a flexible tool for exploring RL strategies that control *in silico* biological processes. We demonstrate PHYSIGYM's potential through a case study on a simplified tumor microenvironment (TME) toy model, in which a Soft Actor-Critic agent learns targeted drug-delivery policies that steer the microenvironment toward an anti-tumoral state and, in the best episodes, achieve tumor elimination. We further use this case study to systematically compare a family of observation spaces from a partially observed macrophage-count baseline, through spatial scalar summaries, to full multi-channel spatial images. Training on a structured rectangle layout and testing on a held-out network-field layout, we find that spatially explicit image observations generalize markedly better than scalar summaries: image modes reach a mean held-out test return of $\approx$+30 to +32, whereas spatial-scalar summaries train comparably (+43) but collapse to $\approx$0 out of distribution, and the macrophage-only POMDP baseline fails to learn at all (-32). This demonstrates that PHYSIGYM's flexible state-space interface directly impacts policy quality and generalization in spatially structured environments. PHYSIGYM is open-source at `https://anonymous.4open.science/r/PhysiGym-76A7`.

1 Introduction

Biological systems are complex, dynamic, and often difficult to study in real life. In silico modeling aims to mimic these systems, providing a controlled environment for exploring specific aspects of biological processes and testing hypotheses. In particular, agent-based models (ABMs) provide a powerful framework for simulating biological systems where agents (e.g. cells) interact dynamically based on predefined rules. These models are used in oncology and immunology to study emerging behaviors in complex biological systems, with the potential to propose novel therapeutic strategies [Johnson et al., 2025, Laubenbacher et al., 2024, Rocha et al., 2024].

Ordinary differential equations (ODEs) and agent-based models (ABMs) are widely used to simulate the tumor microenvironment (TME). ODEs provide compact system-level descriptions, while ABMs capture spatial and cellular heterogeneity at higher computational cost. For dynamic treatment regimes (DTRs) [Gray and Pallavicini, 1982], however, simulation alone is insufficient: designing adaptive, sequential interventions requires a control framework. Reinforcement learning (RL), where an RL agent learns policies through interaction with an environment, offers such a framework [Barto et al., 1989]. By modeling treatment regimes as a sequence of decisions, RL enables adaptive control of biological simulations [Chakraborty and Murphy, 2014]. For efficient experimentation and policy learning, RL further requires a standardized interface. To address this, we introduce PhysiGym, an interface that connects PhysiCell [Ghaffarizadeh et al., 2018], an ABM framework for multicellular simulations of tissues or cell cultures (tumor microenvironment, other disease tissues, or bacteria colonies), with Gymnasium [Towers et al., 2024], a widely adopted RL application programming interface (API). PhysiCell enables flexible multiscale modeling of biological systems, coupled with a sophisticated underlying physics simulator (the BioFVM diffusion transport solver [Ghaffarizadeh et al., 2016]), while Gymnasium provides a structured framework for training and evaluating RL policies.

The first implementation of RL on an ABM framework based on BioFVM was proposed by Zade et al. [2020], which applied Q-learning [Watkins and Dayan, 1992] to optimize Temozolomide treatment regimes for patients with glioblastoma multiforme, a highly specific problem. More recently, Aif et al. [2025] used PhysiCell as a complex (high-fidelity) simulation environment for tumor growth, combined with a simple (low-fidelity) RL training environment of coupled ordinary differential equations (ODE) to learn treatment strategies for solid tumors.

PhysiGym bridges PhysiCell and Gymnasium to provide a reproducible and scalable framework for integrating agent-based simulations of biological systems with RL strategies. In the following, we present PhysiGym's motivation and design, demonstrating how it directly connects ABMs and RL to control complex biological systems. In particular, we focus on the tumor microenvironment, a biological system made up of cancer cells and surrounding immune and stromal cells that engage in complex interactions and cross-talks, which is proven to be crucial for an effective response to immunotherapy, an important therapeutic approach in an increasing number of types of cancer.

To illustrate the framework, we build a simplified TME toy model and use it as an end-to-end RL case study. This model also serves as a controlled testbed for a practical question that arises in any spatially explicit biological environment: among the many observation spaces PhysiGym can expose (scalars, images, graphs), which ones help an RL agent learn an effective dynamic treatment regime? We find that image-based observations that preserve the spatial layout of cells and substrates outperform compact scalar summaries, both in cumulative return and in generalization to held-out spatial configurations. This is an application-level finding on a deliberately simplified model; the primary contribution of this work is PhysiGym itself.

Contributions. (1) PhysiGym: an open-source framework that bridges the PhysiCell ABM and the Gymnasium RL API, turning any PhysiCell model into an RL environment with minimal code changes. (2) A simplified TME toy model demonstrating an end-to-end RL case study in which an agent learns a spatial drug-delivery policy that drives the microenvironment toward an anti-tumoral state. (3) A systematic comparison of observation spaces on the toy model, showing that spatially explicit (image) representations help RL learn and generalize better than scalar summaries.

2 Methods

2.1 PhysiGym design and implementation

PhysiCell is an open-source ABM framework written in C++ that models multicellular systems using classical mechanics and the BioFVM diffusive transport solver [Ghaffarizadeh et al., 2018, 2016]. Cells are agents governed by type-specific rule sets; substrates such as oxygen or cytokines diffuse through the domain. Intracellular signalling networks [Letort et al., 2018, Ponce-de Leon et al., 2023, Ruscone et al., 2024] and extracellular matrix fibres [Noël et al., 2024] can be added, yielding ABMs that are spatially explicit in 2D or 3D, off-lattice, and multiscale [Metzcar et al., 2019].

RL frames sequential decision-making as a Markov Decision Process (MDP) [Bellman, 1957, Puterman, 2014] with state space S, action space A, transition model T, and reward R. The objective is

$$\arg\max_{\pi \in \Pi} \mathbb{E}\left[\sum_{t=0}^{\tau} \gamma^t r_t \mid s_0,\, \pi\right].$$

Gymnasium [Towers et al., 2024] is the standard Python interface for MDPs, promoting modularity and code reuse across algorithms and environments.

PHYSIGYM connects these two ecosystems by extending the Python interpreter [van Rossum, 1995, Python Software Foundation, 2024] with PHYSICELL. The C++ simulation loop is split into three Python-callable functions: `physicell_start`, `physicell_step`, and `physicell_stop`. For most models only `physicell_step` needs to be modified (e.g. to inject a drug). Three read functions `get_cell`, `get_microenv`, and `get_graph` return cell positions, substrate concentration fields, and cell neighbourhood graphs respectively; all are implemented in `physicellmodule.cpp`.

The interface between PHYSICELL and Gymnasium uses PHYSICELL's built-in *user parameter* and *custom data* constructs, configurable through PhysiCell Studio [Heiland et al., 2024] without recompilation. Users subclass `CorePhysiCellEnv` in `physicell_model.py` to specify observation space, action space, and reward; the base class handles all model-independent Gymnasium bookkeeping. Any existing PHYSICELL model can therefore become a PHYSIGYM environment with minimal code changes. PHYSIGYM is tested on Linux, WSL, and macOS, with continuous integration via GitHub Actions. Documentation and tutorials are at `https://anonymous.4open.science/r/PhysiGym-76A7/man`.

2.2 Implementation of a Tumor Microenvironment Toy Model

We developed a simplified tumor microenvironment (TME) *toy model* to illustrate the potential uses of PHYSIGYM. The model is deliberately small and computationally cheap so that we can train RL agents to convergence many times over (across observation spaces and seeds), while still retaining the spatial, multicellular, and chemically coupled structure that distinguishes agent-based models (ABMs) from ODE surrogates. It includes three cell types, five diffusible substrates, and a set of simple agent-based rules. This toy model highlights how the RL framework can control tumor progression by modulating the whole microenvironment, rather than solely targeting cancer-cell killing.

Domain and Dynamics. The continuous-space environment is simulated on a $64 \times 64\,\mu$m 2D domain. The RL gym step interval is $\Delta t = 15$ min, with episodes running up to 10,080 min (7 simulated days).

Entity Dynamics & The Control Challenge. The environment formulates a highly non-linear, indirect control problem. The agent's action (deploying a localized therapeutic, `drug_1`) has zero direct cytotoxicity. Instead, the agent must learn to manipulate the environment's underlying transition dynamics by reversing a spatial immunosuppression loop. The system comprises three interacting cellular populations whose movements are governed by biochemical spatial gradients (chemotaxis):

- **Tumor Cells (The Target):** Immobile entities that continuously divide. They actively modify the environment by secreting a chemical signal (`tumor_molecule`) that suppresses the local immune system.
- **Macrophages (The Mediators):** Mobile, immortal entities that navigate toward the tumor. They exhibit state-dependent behavior: the tumor's chemical signal successfully "hijacks" them, transforming them from an anti-tumoral to a pro-tumoral state.
- **T-cells (The Effectors):** Mobile, immortal entities capable of triggering tumor apoptosis. However, their spatial navigation is strongly repelled by the presence of pro-tumoral macrophages.

This setup creates a hostile, self-reinforcing feedback loop: the tumor protects itself by converting macrophages, which in turn dynamically block T-cells from reaching the target. As illustrated in Figure 1, the RL agent must maximize reward by learning to precisely deploy its drug to "re-program" the macrophages back to an anti-tumoral state. This spatial intervention remodels the chemical gradients, clearing a path for T-cells to navigate to the tumor and execute their killing function.

2.3 Implementation of an RL approach to control the TME

Our overall objective is to achieve complete tumor eradication while minimizing drug exposure. We describe the action space A, the reward R, and the training and test distributions. The observation space S is the component we vary across experiments: in Section 3 we compare a family of observation modes spanning a partially observed scalar baseline, spatial scalar summaries, and

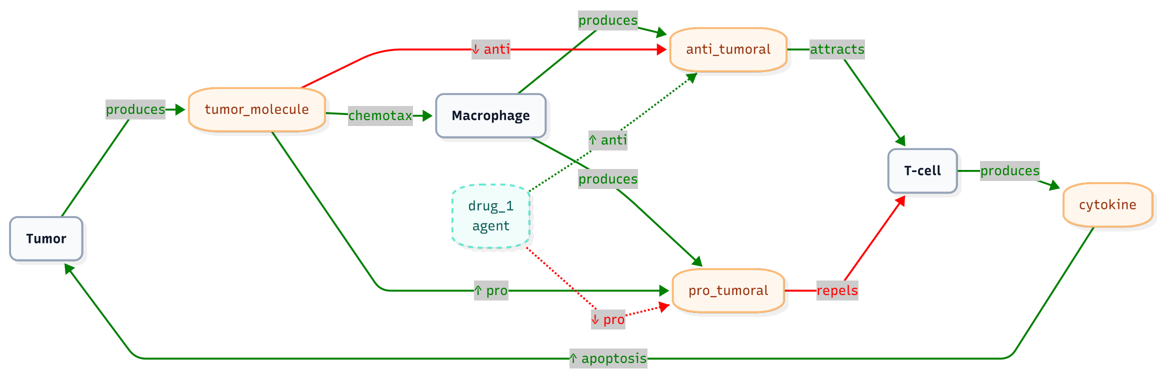

Figure 1: Tumor Microenvironment (TME) signalling network. Green solid arrows denote production, activation, or attraction, while red solid arrows indicate suppression or repulsion. The dashed pathways represent the intervention of the agent-controlled drug (`drug_1`), highlighting its role as an immune-modulator that repolarizes macrophages to indirectly drive T-cell mediated tumor apoptosis.

multi-channel spatial images to assess whether spatially explicit representations help the agent learn and generalize on this toy model.

Action space. At each gym step the agent selects $a_t = (\text{dose}, x, y, r) \in [0, 1]^4$, where dose is the drug concentration applied, (x, y) is the injection centre (normalised domain coordinates), and r is the injection radius (normalised by the domain diagonal).

Reward. The reward is a normalised tumor-suppression signal minus a weighted drug-cost penalty:

$$r_t \;=\; w_{\text{cell}} \, \frac{c_{t-1} - c_t}{c_{t-1}(e^{\lambda \Delta t} - 1)} \;-\; w_{\text{dose}} \, d_t, \tag{1}$$

where c_t is the alive tumor-cell count at step t, λ is the intrinsic growth rate (from `PhysiCell_settings.xml`), d_t is the normalised dose spent, $w_{\text{cell}} = 0.3$, and $w_{\text{dose}} = 2$. Raising the dose weight from 1 to 2 relative to earlier experiments strengthens the pressure to spend drug sparingly, so the agent is rewarded for converting each unit of dose into effective macrophage repolarisation rather than for blanket dosing. The d_t term penalises drug use directly, keeping the learning signal clean without conflating temporal dosing strategy with total exposure.

Training and Test Distributions

To ensure the model learns robust policies rather than overfitting to specific spatial configurations, we train and test on two *structurally distinct* spatial distributions, so that test performance measures out-of-distribution (OOD) generalisation rather than memorisation of the training geometry:

- **Rectangle (training)**: Each cell type is placed uniformly in a narrow vertical strip, with non-overlapping strip positions (Appendix A). This is a highly structured, low-entropy layout.
- **Network-field (held-out test)**: Initial positions are sampled from a spatially correlated random field (Appendix A), producing irregular, clustered configurations never seen during training.

Training on the simple, regular rectangle layout and testing on the richer, irregular network-field is a demanding transfer: a policy that merely exploits the fixed strip geometry of the rectangle cannot solve the network-field, so the test return isolates whether an observation space encodes *transferable* structure. Figure 6 illustrates representative examples of both distributions.

2.4 State space S: observation spaces compared

In a tumor microenvironment there are different cell types and different molecules/chemical compounds (substrates). The simulator can expose several kinds of data: some carry explicit spatial information, others are spaceless summaries (e.g. a small vector of global statistics). Because PHYSI-GYM can build all of these from the same underlying simulation, we use the toy model to study how the choice of observation space affects the dynamic treatment regime that RL learns, comparing nine modes on cumulative return and on generalization to unseen spatial configurations.

We compare observation modes spanning a spectrum from a partially observed scalar baseline to full multi-channel spatial images. Let $N_c = 3$ (tumor, T-cell, macrophage) and $N_s = 5$ substrates. Several modes optionally split the macrophage population into its *M1* (anti-tumoral) and *M2* (pro-tumoral) polarisation states, which we denote with the suffix `m1m2`; the classification rule is given below.

Macrophage polarisation (M1/M2). Macrophages do not carry an explicit polarisation label in the simulator; their state is implicit in the local chemical environment. We recover it from the microenvironment: for each alive macrophage we look up the two competing substrate fields at its nearest voxel and classify it as *M2* (pro-tumoral) if $\mathrm{conc}_{\texttt{pro\_tumoral}} > \mathrm{conc}_{\texttt{anti\_tumoral}}$ at that voxel, and *M1* (anti-tumoral) otherwise. This makes the pro-/anti-tumoral balance the very quantity the agent must flip observable as a count (`scalars` modes) or as two spatial channels (`img` modes), rather than leaving it latent in the substrate fields.

Scalar Observation Spaces

`scalars_macrophages` **(POMDP baseline):** Alive count per cell type, with the macrophage slot split into M1 and M2 counts; each is normalised to the initial population ($s = c/c_{\text{init}} - 1$). This mode exposes only *how many* macrophages are polarised each way and carries no spatial information at all neither cell positions nor substrate maps. It is therefore a strongly partially observed (POMDP) baseline: the true state that governs the transition dynamics (where cells and gradients are) is hidden, and the agent must act on an aggressively compressed summary.
Space: $\mathbb{R}^{N_c+1}$

`spatial_scalars_cells_substrates`**:** Per-type normalised cell counts and per-substrate maxima, augmented with 6 k-means spatial statistics per cell type (cluster occupancy and centroids), giving a compact spatial summary without full images.
Space: $\mathbb{R}^{N_c+N_s+6kN_c}$

`spatial_scalars_cells_m1m2` / `..._substrates_m1m2`**:** As above but with the macrophage population split into M1/M2 for both the scalar counts and the k-means spatial features (`..._substrates_m1m2` additionally keeps the substrate maxima).
Space: $\mathbb{R}^{(N_c+1)+[N_s]+6k(N_c+1)}$

`spatial_scalars_cells_spatial_no_scalars_substrates_m1m2`**:** M1/M2 cell counts plus k-means spatial features for *both* cell types and substrate fields (spatially resolved substrate summaries in place of scalar maxima).
Space: $\mathbb{R}^{(N_c+1)+6kN_s+6k(N_c+1)}$

Image-Based Observation Spaces

`img_mc_cells` **(I1):** One 64×64 channel per cell type.
Space: $\mathbb{R}^{N_c\times64\times64}$ (`uint8`)

`img_mc_cells_m1m2`**:** I1 with the macrophage channel split into separate M1 and M2 spatial maps.
Space: $\mathbb{R}^{(N_c+1)\times64\times64}$ (`uint8`)

`img_mc_cells_substrates` **(I2):** I1 concatenated with all five substrate concentration maps, providing direct spatial feedback on drug distribution and the polarising factors.
Space: $\mathbb{R}^{(N_c+N_s)\times64\times64}$ (`uint8`)

`img_mc_cells_substrates_m1m2` **(full information):** I2 with the additional M1/M2 macrophage channels. This mode carries every signal the others expose cell positions, substrate fields,

and the explicit polarisation split and serves as the near-fully-observed upper reference against which the compressed modes are compared.
Space: $\mathbb{R}^{(N_c+N_s+2)\times 64\times 64}$ (`uint8`)

An observability spectrum

Together these modes trace an observability spectrum, from `scalars_macrophages` a POMDP that sees only polarisation counts to `img_mc_cells_substrates_m1m2`, which spatially resolves every state variable relevant to the control problem. The indirect drug mechanism (Section 2.2) requires multi-step credit assignment: the agent injects drug, macrophages repolarise over several steps, anti-tumoral gradients shift, T-cells migrate, and only then do tumor cells die. Scalar summaries cannot tell the agent whether an injection reached macrophages, whether polarisation occurred, or where T-cells will concentrate next; only the per-pixel substrate and M1/M2 channels of the image modes make the polarisation state and drug distribution spatially legible at each step.

3 Results

3.1 Aggregated performance

Table 1 summarises final performance (seeds averaged). Figures 2 and 4 show learning curves vs. cumulative environment steps with mean $\pm$ std shading across seeds.

Table 1: Performance at end of training (train = rectangle, test = network-field, seeds averaged). Train$_\mu$/Test$_\mu$: mean EWMA-50 return. Train$_\sigma$/Test$_\sigma$: std across seeds; n = number of seeds. This is the curated main set; the remaining `m1m2` scalar variants appear in Appendix C, and 95% bootstrap confidence intervals on the mean in Appendix D. RAND is a random-action policy (5 seeds), included as a learning-free reference: modes below it fail to beat random behaviour.

ID	**Mode**	n	**Train$_\mu$**	**Train$_\sigma$**	**Test$_\mu$**	**Test$_\sigma$**
I2m	`img_mc_cells_substrates_m1m2`	5	+40.8	8.5	**+32.5**	9.4
I1	`img_mc_cells`	6	+30.5	8.7	+32.4	10.5
I2	`img_mc_cells_substrates`	7	+42.5	6.6	+30.4	7.4
S3s	`spatial_scalars_cells_substrates`	5	+42.9	18.8	-0.0	9.1
RAND	random policy (baseline)	5	-18.4	4.0	-25.3	3.0
POMDP	`scalars_macrophages`	5	-20.4	8.3	-31.8	2.3

3.2 Key findings

Generalization, not training fit, separates the modes. On the training rectangle layout the spatial-scalar summary S3s is fully competitive its training return ($+42.9$) matches the best image mode I2 ($+42.5$). The distinction appears only out of distribution: on the held-out network-field, the three image modes retain $\approx$+30 to $+32$ test return, while S3s collapses to -0.0. Training fit alone would have ranked the scalar summary among the best; only the OOD test reveals that it has overfit the fixed strip geometry of the rectangle. This separation is statistically supported: the image modes' 95% bootstrap confidence intervals on the mean test return lie entirely above $+24$ and do not overlap those of any spatial-scalar mode, all of which contain zero (Appendix D).

Spatially explicit observations transfer; scalar summaries do not. All three image modes `img_mc_cells` (I1, $+32.4$), `img_mc_cells_substrates` (I2, $+30.4$), and the full-information `img_mc_cells_substrates_m1m2` (I2m, $+32.5$) generalize to the network-field, and the gaps among them are within one seed-standard-deviation. That I1 already generalizes well indicates that, in this toy model, preserving the *spatial layout* of cells is the primary driver of transfer; the extra substrate and M1/M2 channels of I2/I2m do not further separate on average, though I2m gives the single best test return.

The POMDP baseline is no better than acting randomly. `scalars_macrophages`, which observes only M1/M2 counts and no spatial information, fails to improve on either distribution

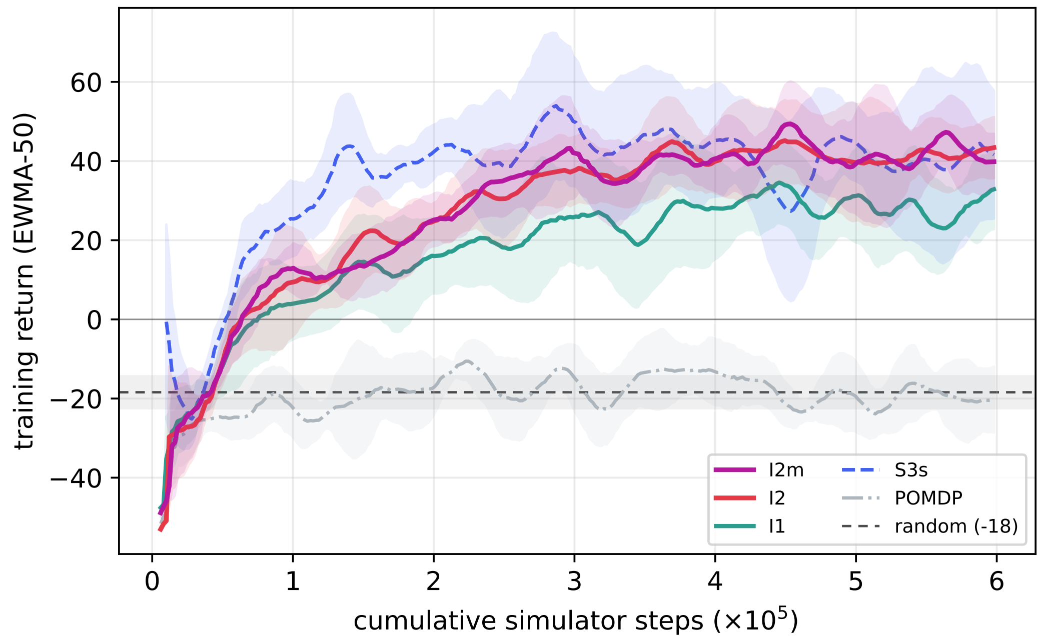

Figure 2: Training return on the rectangle layout (EWMA-50, mean $\pm 1\sigma$ across seeds) vs. cumulative *simulator* steps ($\times 10^5$; each of the 10^5 agent decisions is held for 6 simulator steps, giving a 6×10^5 budget). All modes learn on the training layout except the `scalars_macrophages` POMDP baseline, which flatlines near -20. Spatial-scalar summaries (S3s) train as fast as, or faster than, the image modes here the separation only appears out of distribution (Figure 4).

(train -20.4, test -31.8) and, crucially, does *not* beat a random-action policy (train -18.4, test -25.3; RAND in Table 1). Its test return is if anything below random, and the two bootstrap CIs barely overlap (Appendix D): the macrophage-count signal is not merely uninformative but actively misleading, steering dosing worse than chance. Knowing *how many* macrophages are polarised without knowing *where* the drug, cells, and gradients are is insufficient to solve the indirect, spatially mediated control problem a concrete illustration of how observation-space choice can render an otherwise learnable task effectively unlearnable. By contrast, every spatial-scalar mode, though it fails to generalize, still clears the random baseline on test (≥ 0 vs. -25.3), confirming they learn *some* transferable structure just far less than the image modes.

On seed robustness. Unlike a cleaner earlier run, we do not observe a systematic seed-variance advantage for any family on this benchmark: end-of-training test std is 7–12 for the image modes and 9 for S3s (Table 1, Figure 3). The POMDP baseline's low std (2.3) reflects consistent failure rather than robustness.

3.3 The role of the M1/M2 split and spatial scalars

The full mode family (Appendix C, Table 3) lets us isolate the effect of the explicit M1/M2 polarisation split. Adding it to the cell image (I1→`img_mc_cells_m1m2`) does not improve average OOD test return ($+32.4 \rightarrow +26.0$), and adding it on top of the full image (I2→I2m) leaves it essentially unchanged ($+30.4 \rightarrow +32.5$): once the substrate fields are spatially available, the CNN can already read out polarisation from the `pro_tumoral`/`anti_tumoral` channels, so the pre-computed split is redundant. Among the spatial-scalar modes, none generalize (all within $[-0.0, +9.5]$ test return) regardless of whether they carry the M1/M2 split or spatially resolved substrate summaries the common failure is the loss of per-cell spatial layout, which the k-means summaries do not preserve.

Appendix E shows a matched example episode comparing an image mode, a spatial-scalar mode, and the POMDP baseline rolled out from an identical network-field initial condition, illustrating the failure mode that summarises the comparison: without a spatially explicit representation the agent

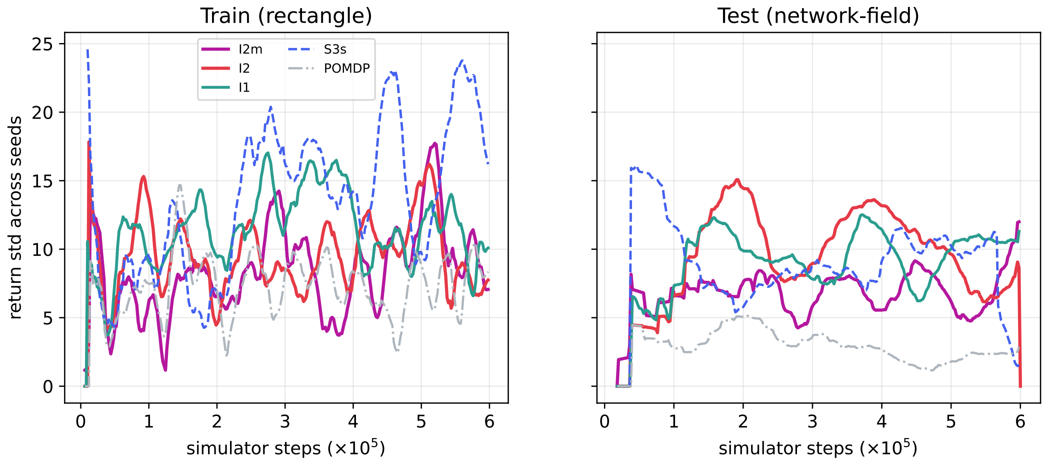

Figure 3: Standard deviation of return across seeds vs. simulator steps: training on rectangle (left) and test on network-field (right). Inter-seed variability is broadly comparable across the learning modes and fluctuates over training, so we do not claim a systematic variance advantage for any single family; the POMDP baseline has the lowest test variance simply because it converges to a uniformly poor policy. At end of training the image modes reach test return $\approx$+30 with std $\approx$7–12 (Table 1).

cannot target the drug precisely enough to convert dose into effective macrophage repolarisation on an unseen layout the image mode reaches near-eradication (3 tumor cells) while the scalar and POMDP modes leave 70 and 130 cells respectively.

4 Discussion

The central outcome of this work is PhysiGym itself: a framework that lets RL agents control a full agent-based TME through a standardised interface, without recompilation, and with arbitrary observation and action spaces. The toy-model case study shows the framework end-to-end an agent learns a spatial drug-delivery policy that steers the microenvironment toward an anti-tumoral state and, in the best episodes, eradicates the tumor.

The toy model also produces a clear result on observation spaces, and crucially one that only surfaces under distribution shift. On the training rectangle layout, spatial-scalar summaries are fully competitive with image observations; it is on the held-out network-field that spatially explicit (image) observations pull ahead, generalizing to $\approx$+30 test return while the scalar summaries collapse to $\approx$0. The CNN in the image modes can compute cross-channel co-localisation statistics (e.g. whether `drug_1` and macrophages overlap at the same location) directly from the pixel layout statistics that transfer across geometries whereas scalar modes that encode k-means cluster centroids and absolute positions overfit the fixed strip geometry of the training layout. Notably, once cell layout is spatially available, preserving the substrate fields (I2) or pre-computing the M1/M2 split (I2m) does not further improve average transfer over the cell-only image (I1) on this toy model, though the full-information mode gives the single best test return.

The overfit is a drug-aiming failure, measurable in the actions. The claim above is not merely architectural: it is visible directly in the policies' behaviour. For every logged episode we compute the mean distance between the drug-injection centre (`action_x`, `action_y`) and the tumour centroid (from the episode initial conditions), averaged over the steps on which a dose is actually applied, and compare this aiming error between the training rectangle and the held-out network-field (Fig. 5). On the training layout all six modes aim equally well (mean normalised error $\approx$0.34–0.36 for both families). Under the geometry shift, image modes hold their aim or slightly improve it (I2: $0.359 \rightarrow 0.343$; I2m: $0.352 \rightarrow 0.333$), whereas every scalar mode's aim *degrades* (S3s: $0.341 \rightarrow 0.414$; S3m: $0.348 \rightarrow 0.406$) and its step-to-step variance roughly doubles. In other words, the scalar policies learn *where* the rectangle tumour tends to sit rather than *how to find* the tumour, so when the network-field moves it they inject into empty tissue; the median final tumour count on test rises accordingly (image

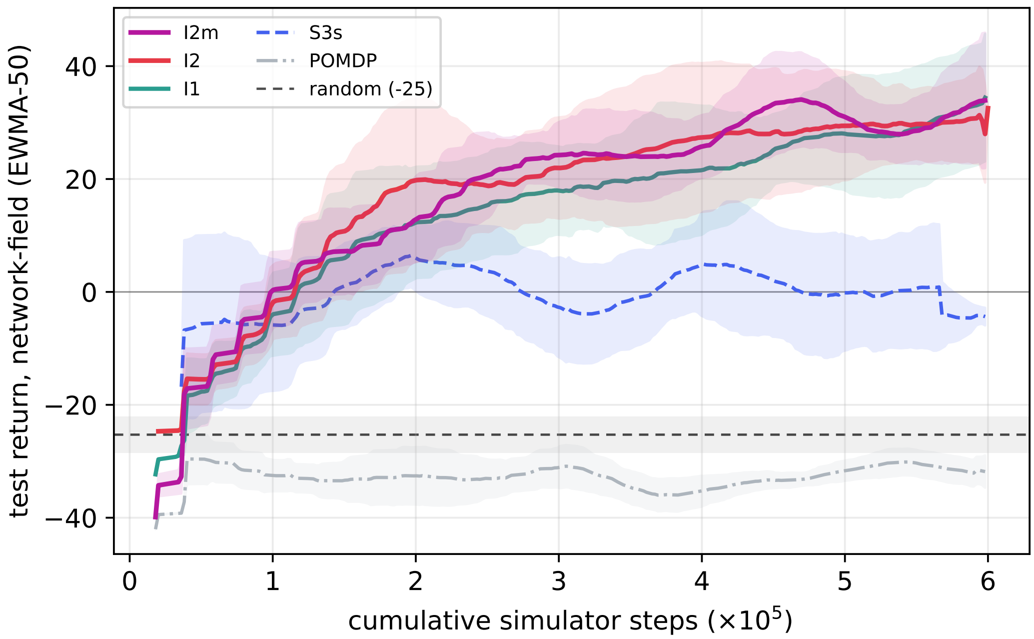

Figure 4: Held-out test return on the network-field layout (mean $\pm 1\sigma$ across seeds, EWMA-50) vs. cumulative simulator steps. The image modes (I2m, I2, I1) climb steadily to $\approx$+30, whereas the spatial-scalar summary (S3s) which trained *best* (Figure 2) collapses to $\approx$0, revealing that it overfits the fixed rectangle geometry rather than learning transferable structure. The dashed line marks the random-policy baseline (test ≈ -25, shaded $\pm 1\sigma$); the `scalars_macrophages` POMDP baseline tracks at or below it throughout, i.e. never beats random. Generalization, not training fit, is where spatially explicit observations separate from scalar summaries.

$\approx$73–99 vs. scalar 107–136). This is the concrete mechanism behind the aggregate return gap: the generalisation failure is an aiming failure induced by position-absolute features.

We emphasise that this is an application-level observation on a deliberately simplified model, not a universal claim about biological control. In more complex or realistic models or with different reward designs, action spaces, or RL algorithms the relative value of cell versus substrate, and scalar versus image, may differ. PhysiGym is precisely the tool that makes such questions answerable in a reproducible way.

Limitations. The toy model is intentionally small (64 μm, 2D) and cheap to simulate. Scaling to 3D or larger domains will increase the cost of image observations. All experiments use a single RL algorithm (SAC); alternative algorithms or model-based methods may interact differently with observation spaces. Each mode is trained with a modest number of seeds (n = 5–7), unevenly distributed across modes; we mitigate this with bootstrap confidence intervals (Appendix D) rather than relying on raw seed-standard-deviation, but the conclusions would be strengthened by more seeds and by testing transfer in both directions between layouts. We also study a single train/test layout pair (rectangle→network-field); the reverse direction is left to future work. Complex biological models and/or 3D models are left to future work.

5 Conclusion

We presented PhysiGym, an open-source framework that bridges the PhysiCell ABM with Gymnasium RL, enabling direct policy learning in spatially explicit biological simulations. Using a TME model with an indirect drug mechanism, we showed that spatially explicit image observations are the critical state representation for *generalization*: training on a structured rectangle layout and testing on a held-out network-field, image modes retain $\approx$+30 to +32 test return while spatial-scalar summaries competitive during training collapse to $\approx$0 out of distribution, and a macrophage-count-

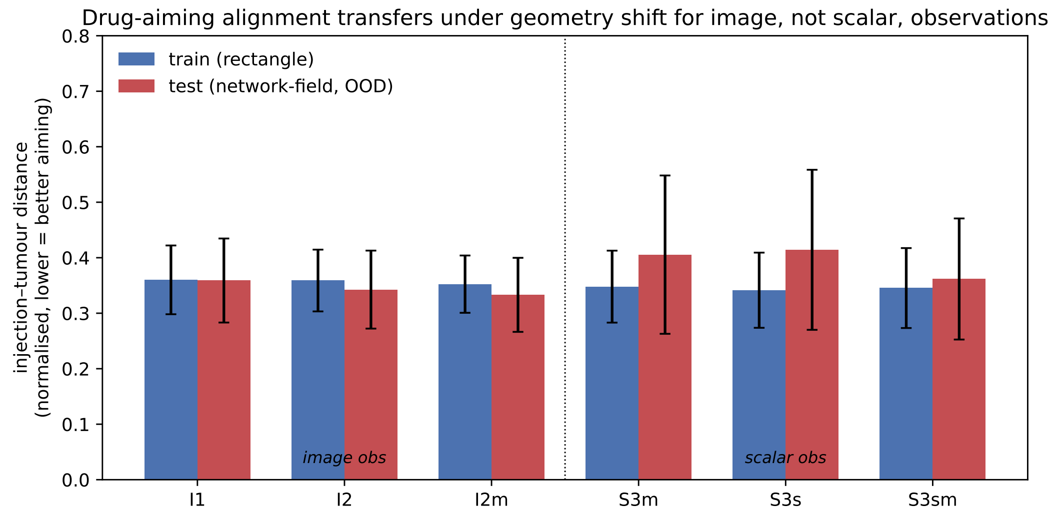

Figure 5: Drug-aiming alignment under geometry shift. Bars show the mean normalised distance between the drug-injection centre and the tumour centroid over dosed steps (lower is better aiming); error bars are the across-episode standard deviation. On the training rectangle (blue) image and scalar observations aim equally well. On the held-out network-field (red) image modes retain their aim while every scalar mode degrades with markedly higher variance, exposing the overfit as a failure to re-target the tumour rather than a loss of dosing ability.

only POMDP baseline fails to learn at all. The gap is a generalization phenomenon: what separates the representations is not training fit but whether the encoded structure transfers across spatial configurations. PhysiGym provides the infrastructure to study such questions in a principled, reproducible manner, and we anticipate it will serve as a testbed for both computational biology and RL communities.

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A Generation of Initial Conditions

Rectangle (training layout). Each cell type is placed uniformly in a vertical strip of width 10% of the x-axis, with non-overlapping strip positions sampled from $[0.1, 0.2]$, $[0.3, 0.5]$, $[0.6, 0.8]$ (shuffled across types). This is a highly regular, low-entropy layout.

Network-field (held-out test layout). White Gaussian noise $\eta(\mathbf{x}) \sim \mathcal{N}(0, 1)$ is convolved with a Gaussian kernel G_σ ($\sigma = \ell_c/\sqrt{2}$, $\ell_c = 45\,\mu\text{m}$) to produce a correlated field F. An additional local noise component $\tilde{\xi}$ filtered at width $\ell_c/3$ is added with small weight α to introduce local heterogeneity. The field is normalised to $[0, 1]$ and cells are sampled from positions with $\hat{F} > 0.55$, with probability proportional to $\hat{F}$, producing irregular, clustered configurations that are structurally distinct from the training rectangle.

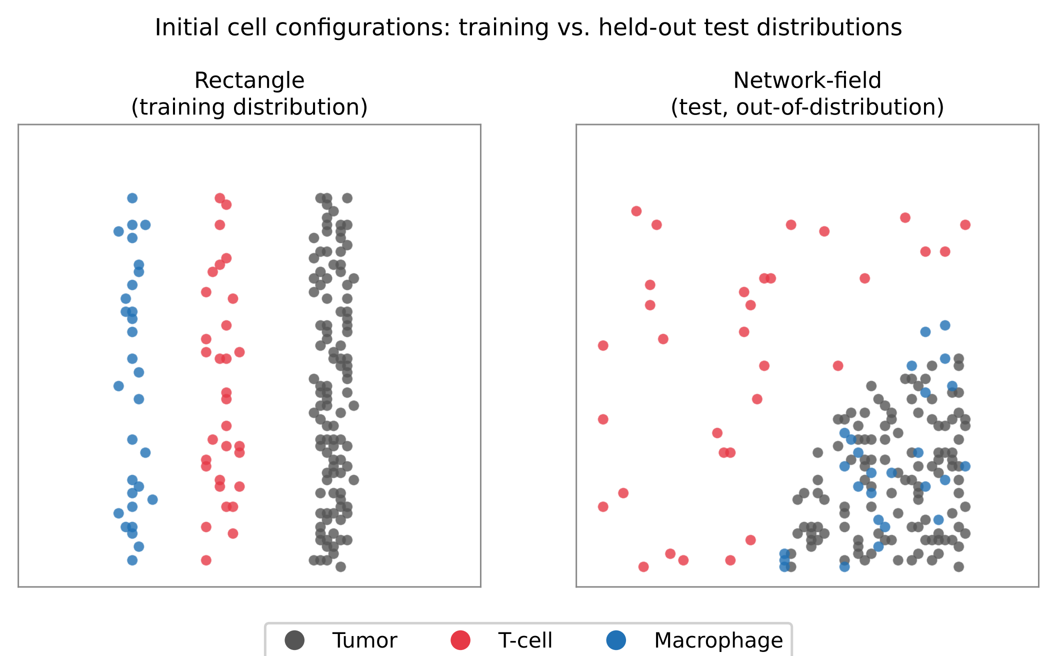

Figure 6: Representative examples of the training (rectangle) and held-out test (network-field) spatial layouts.

B Training algorithm and setup

To address our control objective maximizing cancer-cell eradication while minimizing drug administration we employ Soft Actor-Critic (SAC) [Haarnoja et al., 2018], a state-of-the-art RL algorithm for continuous control. SAC is well suited to our setting due to its high sample efficiency, and as a deep RL method it accommodates different observation types, such as image-based representations, through Convolutional Neural Networks (CNNs) [O'shea and Nash, 2015]. A key challenge is the computational cost of the simulator; to alleviate this bottleneck we adopt an asynchronous parallel training framework [Mnih et al., 2016, Liu et al., 2025]. A GPU-resident learner updates policy and value networks asynchronously while 13 CPU-resident environment processes generate experience in parallel.

Compute. Each run trains a single observation mode for one seed to 10^5 agent decisions (6×10^5 simulator steps) in approximately 5 hours of wall-clock time on a single workstation (one GPU-resident learner with 13 CPU environment workers). The asynchronous design decouples the GPU learner from the comparatively slow PhysiCell simulator, so throughput is limited by the CPU environment workers rather than by gradient updates; this is what makes the full observation-space sweep (nine modes $\times$ 5–7 seeds) tractable on a single machine.

Action repeat. Each action selected by the agent is held fixed for 6 consecutive simulator steps (*action repeat* of 6) before the next decision. This serves two purposes. First, it stabilises learning: with a 15-minute decision interval on a slow, indirect biological process (drug injection $\rightarrow$ macrophage repolarisation $\rightarrow$ gradient shift $\rightarrow$ T-cell migration $\rightarrow$ tumor death), consecutive raw simulator steps are highly correlated and a fresh decision at every step injects high-frequency jitter into the drug field without changing the underlying dynamics; committing to an action for several steps lets its effect materialise before it is revised, which empirically yields smoother dose trajectories and lower-variance value targets. Second, it improves simulator throughput per decision: the 10^5 agent decisions per run correspond to 6×10^5 simulator steps of experience, so the effective interaction budget (and all step axes in Figures 2–4) is 6×10^5. Action repeat is a standard device for temporally extended control and is well motivated here by the slow timescale of the controlled process relative to the simulator step. The specific value AR$=6$ is selected by a sensitivity sweep over AR$\in\{1,\dots,8\}$ (Appendix B.4, Figure 7): the action-smoothness cost drops steeply and plateaus by AR$\approx$6, and the composite control objective reaches its plateau over AR$=$ 5–8, so AR$=6$ is the smallest repeat that attains both.

B.1 Asynchronous SAC

Algorithm 1 Asynchronous SAC with Parallel Environments [Liu et al., 2025]

Require: Config d_{arg}, buffer size $|\mathcal{B}|$, LRs, γ, τ

```
Init shared queues actor_queue, sample_queue; stop signal stop_event
Actor Process (CPU): init N_env = 13 envs, local π_θa
while not stop_event do
    Receive θ from actor_queue if available
    Act: random if t < T_start, else π_θa
    Step envs; push (s, a, r, s', d) to sample_queue
end while
Learner Process (GPU): init π_θ, Q_φ1,2, targets, D
while total samples < T do
    Drain sample_queue into D
    for N_loops gradient steps do
        Sample minibatch; compute SAC target y; update Q, π, α; soft-update targets
    end for
    Push updated θ to actor_queue
end while
```

B.2 Neural Architecture

Scalar modes utilize a Multi-Layer Perceptron (MLP):

$$\mathrm{FC}(256) \xrightarrow{\text{Mish}} \mathrm{FC}(256)$$

Image modes utilize an IMPALA-style CNN [Espeholt et al., 2018]. The feature extraction and projection pipeline is defined as:

$$
\begin{aligned}
&\mathrm{Conv}(n_c \to 32,\ k{=}8{\times}8,\ s{=}4) \xrightarrow{\text{Mish}} \\
&\mathrm{Conv}(32 \to 64,\ k{=}4{\times}4,\ s{=}2) \xrightarrow{\text{Mish}} \\
&\mathrm{Conv}(64 \to 64,\ k{=}3{\times}3,\ s{=}1) \xrightarrow{\text{Mish}} \mathrm{Flatten} \to \mathrm{FC}(256) \to \mathrm{FC}(256)
\end{aligned}
$$

B.3 Hyperparameters

Table 2: SAC hyperparameters (TME v2 experiments).

Parameter	Symbol	Value
Agent decisions	T	1×10^5
Action repeat	–	6
Effective simulator steps	$6T$	6×10^5
Replay buffer size	$\|\mathcal{B}\|$	2×10^5
Batch size	B	832
Warm-up steps	T_{start}	5,000
Policy / critic LR	η_π, η_Q	3×10^{-4}
Discount factor	γ	0.99
Target smoothing	τ	0.005
Reward weights	$w_{\text{cell}}, w_{\text{dose}}$	$0.3,\ 2.0$
Entropy autotuning	–	enabled
Parallel envs	N_{env}	13
Test frequency	–	every 4 episodes

B.4 Choice of action repeat

We select the action-repeat value with a dedicated sensitivity sweep, run independently of the main observation-space experiments. We sweep action_repeat $\in \{1, \ldots, 8\}$ and, for each value, measure three end-of-run quantities aggregated over repeated rollouts (with 95% bootstrap CIs): tumor reduction (control efficacy), an action-smoothness cost $\sum_t \|a_t - a_{t-1}\|^2$ (policy jitter; lower is better), and a composite objective combining tumor reduction, dose, and smoothness (Figure 7).

Two effects drive the choice. (i) *Smoothness*: the action-smoothness cost falls steeply and monotonically with action repeat from $\approx$110 at AR$=1$ to $\approx$17 at AR$=6$ and then flattens, so most of the stabilisation benefit is already captured by AR$=6$ and larger values add little. This matches the physical timescale of the task: with a 15-minute decision interval on a slow immune process, re-deciding every simulator step injects high-frequency drug-field jitter that does not change the underlying dynamics. (ii) *Composite objective*: the objective is worst at AR$=1$ (-0.88) and rises to a plateau over AR$=$ 5–8, with AR$=6$ at -0.07; within that plateau the values are within each other's confidence intervals, so AR$=6$ is chosen as the smallest repeat that reaches the objective plateau while sitting at the knee of the smoothness curve. This makes action repeat a deliberate, evidence-backed choice rather than an incidental hyperparameter.

C Full observation-mode comparison

Table 3 reports all nine observation modes evaluated, including the `m1m2` scalar variants omitted from the curated main-text Table 1. Train and test returns are end-of-training EWMA-50 means with the standard deviation across seeds; train = rectangle, test = network-field.

D Bootstrap confidence intervals

Because each mode is trained with only $n =$ 5–7 seeds, the across-seed standard deviations reported in Tables 1 and 3 are themselves noisy estimates of dispersion. To quantify uncertainty on the *mean* return more robustly, we bootstrap over seeds: for each mode we resample the seeds with replacement

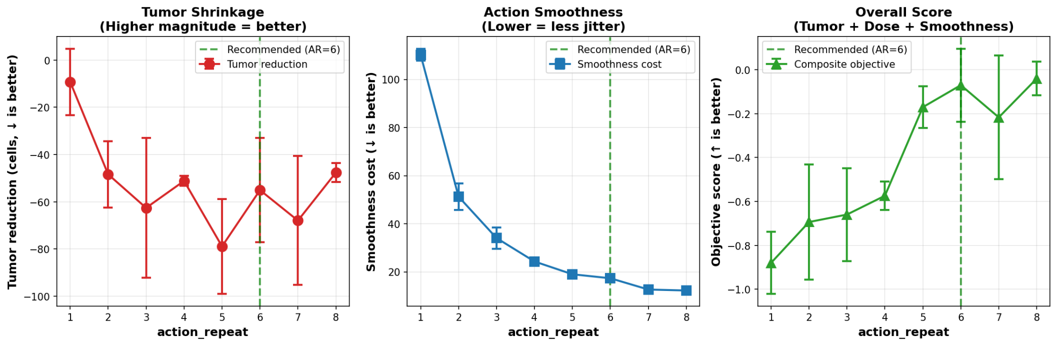

Figure 7: Action-repeat sensitivity sweep (independent of the observation-space experiments; error bars are 95% bootstrap CIs). **Left:** tumor reduction (more negative = more cells removed). **Centre:** action-smoothness cost drops sharply and plateaus by AR≈6. **Right:** composite objective (higher is better) rises to a plateau over AR= 5–8; AR= 6 (dashed) is the smallest repeat on that plateau and at the knee of the smoothness curve.

Table 3: All observation modes (seeds averaged; train = rectangle, test = network-field). Modes are ordered by held-out test return.

ID	Mode	n	Train$_\mu$	Train$_\sigma$	Test$_\mu$	Test$_\sigma$
I2m	`img_mc_cells_substrates_m1m2`	5	+40.8	8.5	+32.5	9.4
I1	`img_mc_cells`	6	+30.5	8.7	+32.4	10.5
I2	`img_mc_cells_substrates`	7	+42.5	6.6	+30.4	7.4
I1m	`img_mc_cells_m1m2`	5	+35.0	13.1	+26.0	12.4
S5m	`spatial_scalars_cells_spatial_no_scalars_substrates_m1m2`	5	+53.6	34.2	+9.5	9.6
S3sm	`spatial_scalars_cells_substrates_m1m2`	5	+38.3	27.8	+2.6	12.1
S3m	`spatial_scalars_cells_m1m2`	6	+45.7	26.8	+2.6	9.7
S3s	`spatial_scalars_cells_substrates`	5	+42.9	18.8	−0.0	9.1
RAND	random policy (baseline)	5	−18.4	4.0	−25.3	3.0
POMDP	`scalars_macrophages`	5	−20.4	8.3	−31.8	2.3

$B = 10{,}000$ times, recompute the mean end-of-training return of each resample, and report the 2.5th–97.5th percentiles of that bootstrap distribution as a 95% confidence interval on the mean (Table 4). The same procedure applied per step yields the shaded bands in Figures 8 and 9.

The bootstrap intervals sharpen the main-text claims. On the held-out network-field, the three image modes have test-return CIs that lie entirely above $+24$, whereas every spatial-scalar mode has a test-return CI that *contains zero* (e.g. S3s $[-6.2, +9.1]$), so their generalization is not statistically distinguishable from a non-learning policy. The image and spatial-scalar CIs do not overlap, confirming that the generalization gap is significant and not an artefact of seed selection. The `scalars_macrophages` POMDP baseline has a tight, strongly negative CI $[-33.4, -29.4]$, reflecting consistent failure rather than robustness and it does not overlap the random policy's test CI $[-28.2, -22.8]$ from the favourable side, so the macrophage-only observation performs *no better than, and plausibly worse than, random dosing*.

E Example Episode Comparisons

Figure 10 shows a single matched test episode comparing an image mode (`img_mc_cells_substrates`), a spatial-scalar mode (`spatial_scalars_cells_substrates`), and the POMDP baseline (`scalars_macrophages`) step by step. To guarantee an exact comparison, all three agents are rolled out (deterministically, using the policy mean action) from the *identical* held-out network-field initial condition the same 124-tumor-cell layout replayed from file so the only difference between the three trajectories is the observation mode.

The episode cleanly separates the three regimes. `img_mc_cells_substrates` drives the tumor to near-eradication (3 cells remaining, $+223$ cumulative return); it spends the most drug

Table 4: Mean end-of-training return with a 95% bootstrap confidence interval on the mean ($B =$ 10,000 seed resamples; train = rectangle, test = network-field). Ordered by test mean.

ID	**Mode**	n	**Train$_\mu$ (95% CI)**	**Test$_\mu$ (95% CI**
I2m	`img_mc_cells_substrates_m1m2`	5	+40.8 [+33.0, +47.3]	+32.5 [+24.4, +4
I1	`img_mc_cells`	6	+30.5 [+23.1, +36.7]	+32.4 [+24.2, +4
I2	`img_mc_cells_substrates`	7	+42.5 [+37.4, +47.2]	+30.4 [+24.5, +3
I1m	`img_mc_cells_m1m2`	5	+35.0 [+22.7, +45.6]	+26.0 [+13.5, +3
S5m	`spatial_scalars_cells_spatial_no_scalars_substrates_m1m2`	5	+53.6 [+25.0, +85.1]	+9.5 [+0.4, +17
S3sm	`spatial_scalars_cells_substrates_m1m2`	5	+38.3 [+20.1, +65.9]	+2.6 [−6.2, +13
S3m	`spatial_scalars_cells_m1m2`	6	+45.7 [+24.9, +67.3]	+2.6 [−5.8, +10
S3s	`spatial_scalars_cells_substrates`	5	+42.9 [+24.7, +58.8]	−0.0 [−6.2, +9.
RAND	random policy (baseline)	5	−18.4 [−21.9, −14.9]	−25.3 [−28.2, −2
POMDP	`scalars_macrophages`	5	−20.4 [−28.0, −13.8]	−31.8 [−33.4, −2

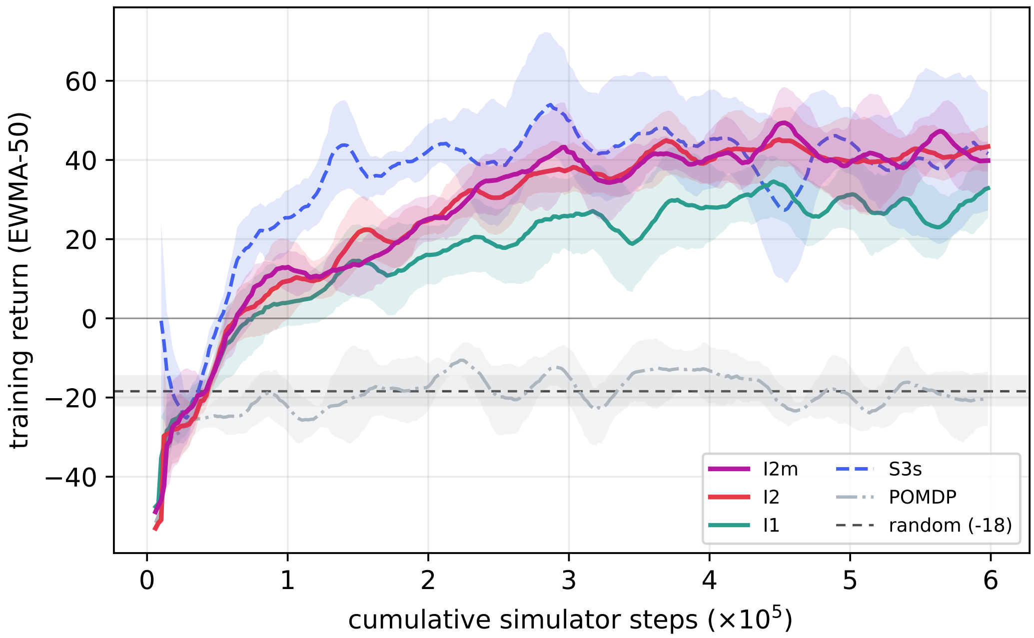

Figure 8: Training return on rectangle (EWMA-50, mean with 95% bootstrap CI band over seeds) vs. cumulative simulator steps. Curated main modes only.

(4.5 total dose) but places it precisely enough visibly dosing in sustained, targeted bursts (Figure 10, bottom) to convert dose into effective macrophage repolarisation and tumor clearance. `spatial_scalars_cells_substrates` achieves partial control (70 cells, +30 return, 2.8 dose): it suppresses growth early but plateaus and cannot finish, lacking the spatial substrate feedback needed to target the final push. `scalars_macrophages` (POMDP) barely acts (0.9 total dose) and the tumor *grows* past its starting size (130 cells, −6.4 return): seeing only macrophage-polarisation counts, it cannot localise where to inject, so it under-doses and fails consistent with its aggregate performance at or below the random baseline.

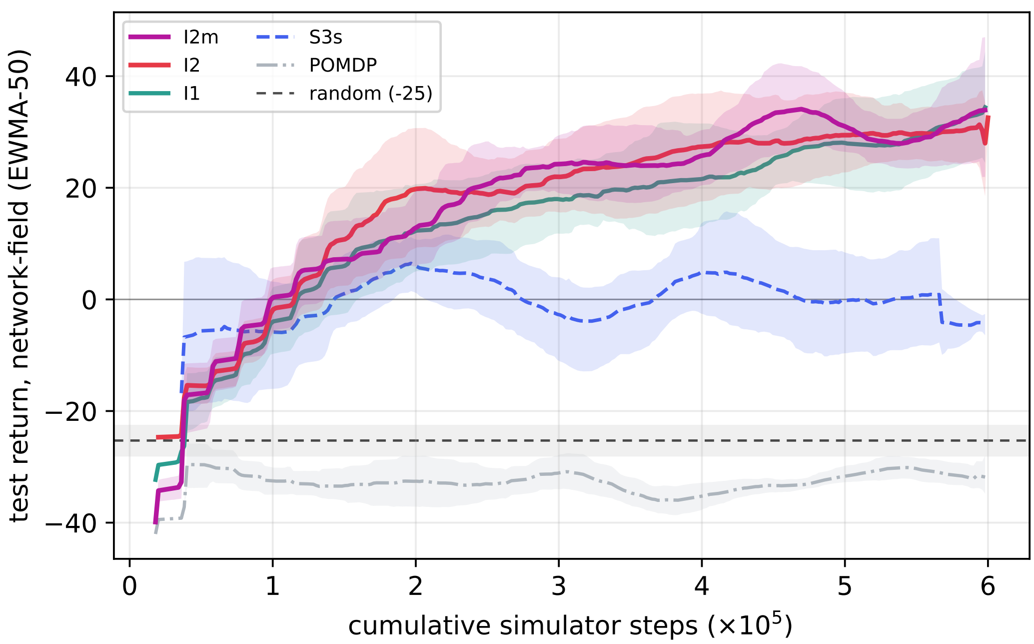

Figure 9: Held-out test return on network-field (EWMA-50, mean with 95% bootstrap CI band over seeds) vs. cumulative simulator steps. The image modes' bands separate from the spatial-scalar summary (S3s), whose band hugs zero, and from the POMDP baseline.

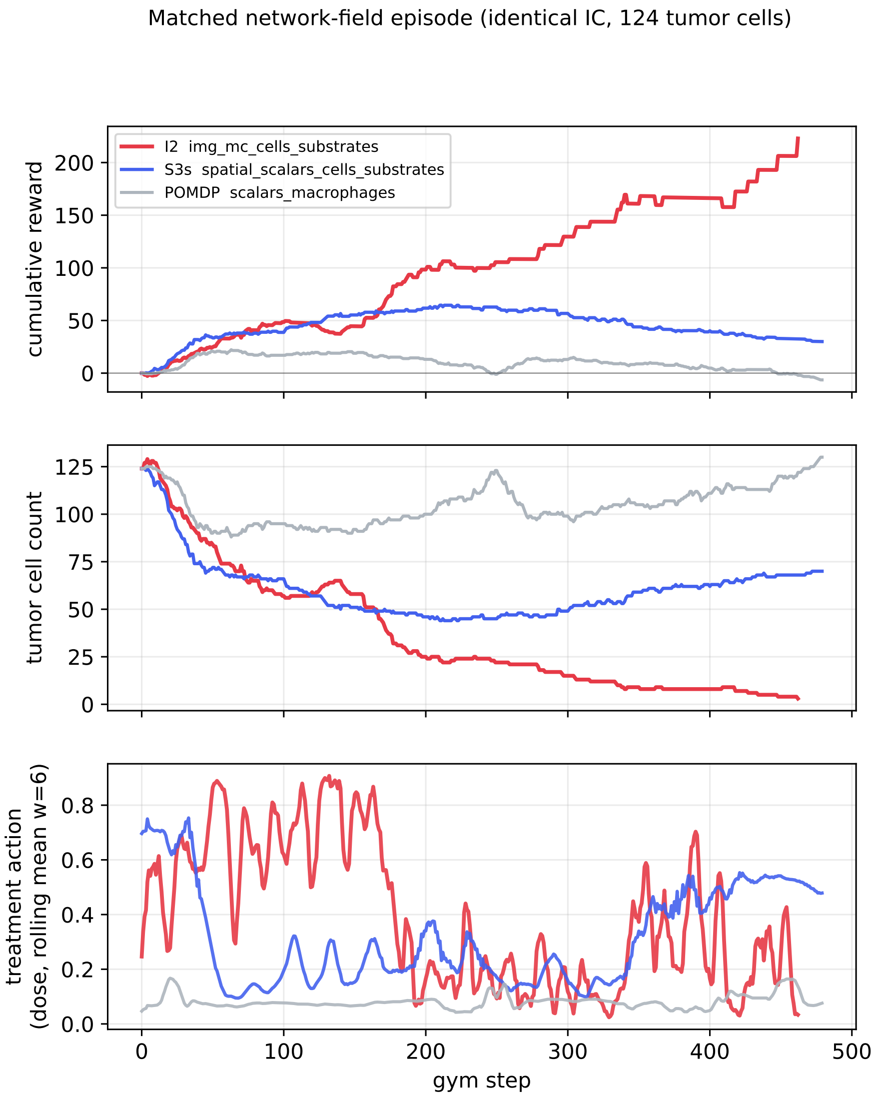

Figure 10: **Matched network-field episode (identical initial condition,** 124 **tumor cells).** Cumulative reward (top), tumor cell count (middle), and the dose action (bottom, rolling mean over 6 steps matching the action-repeat horizon used in training; the raw per-step signal is noisy and unreadable at full resolution) for `img_mc_cells_substrates` (red), `spatial_scalars_cells_substrates` (blue), and the `scalars_macrophages` POMDP baseline (grey), all replayed from the same IC. `img_mc_cells_substrates` reaches near-eradication (3 cells, $+223$ return, 4.5 dose); `spatial_scalars_cells_substrates` achieves partial control (70 cells, $+30$ return, 2.8 dose); the POMDP baseline under-doses and the tumor grows (130 cells, -6.4 return, 0.9 dose). Only the image mode sustains a high, decisive dose action long enough to drive the tumor to near-eradication, whereas `spatial_scalars_cells_substrates` oscillates and eventually lets the tumor regrow. Note that the physical dose delivered to the simulation each step is the concentration action (bottom) multiplied by the voxel volume swept by the independently-controlled injection radius action, so the dose action alone does not linearly predict the physical dose spent.